

Q13

The root mean square speed of the helium atoms is given by $v_{rms} = \sqrt{\frac{3RT}{M_m}}$. Since temperature T and molar mass M_m are constant, the root mean square speed of helium atoms is the same in both sections of the container. Option B seems to be the answer.

The gas pressure is given by $p = \frac{nRT}{V}$. Since V and T are constant but not the number of moles n , the pressure is not the same in both sections.

Since force is proportional to pressure p exerted by the molecules on the wall, the force exerted by molecules in the two sections is not the same. Also, since force F is given by $F = \frac{N(2mv)}{t}$ by N2L, (m and v are the mass and average speed of the helium atom) the rate of collisions $\frac{N}{t}$ of the wall by the atoms in both sections will not be the same.

The Ideal Gas equation can also be written as $pV = \frac{M}{M_m} RT$ or $p = \frac{\rho}{M_m} RT$ where M_m and ρ are the molar mass and density of helium respectively. Since T is constant but not pressure p , the gas density ρ is also not the same in both sections.

Q14

Ans: B